

YOGANSH SHARMA

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🎓 EDUCATION

Indian Institute of Technology Kharagpur

2022 – Present

Dual Degree (B.S. + M.S.) in Exploration Geophysics, expected 2027

🌟 CERTIFICATIONS

Programming Bitcoin by Jimmy Song (Bitshala)

Dec 2024

- Completed an advanced, hands-on cohort exploring Bitcoin protocol fundamentals, cryptography, scripting, and transaction mechanics.
- Built Bitcoin tools from scratch in Python, gaining practical experience with Bitcoin's core architecture.
- Engaged in weekly live sessions, group discussions, and problem-solving exercises with a global developer community.

👥 EXPERIENCE

Kharagpur Open Source Society

Core Team Member

Jun 2023 - May 2024

Executive

May 2024 - Present

- Facilitated training workshops on various technologies including **Git**, **Github**, and **Linux** for over 2,000+ students.
- Served as a speaker in **Golang** Workshop.
- Contributed to rewriting the website using the **Astro** web framework.
- Helped organize Kharagpur Winter of Code 2023, inspired by Google Summer of Code, which had 69 projects and over 1400 students.
- Promoted and advocated for using **open-source** software amongst the student body.

Technology Students Gymkhana

Secretary Web

Aug 2023 – Present

- Maintain and update the Gymkhana website, utilizing **ReactJS** and **AWS** for hosting.
- Collaborate with other office bearers to implement and execute proposals, enhancing overall efficiency and productivity.
- Led the revamp of the certificate generator, resulting in streamlined processes and improved user satisfaction.

git PROJECTS

KWoC Backend

Jun 2023 – Present

- Rewrote the **RESTful API** using **Golang** with the mux router, enhancing performance and maintainability.
- Implemented **Continuous Integration (CI)** to automate documentation generation with **Swagger** and enforce code quality with pre-commit hooks.
- Orchestrated **Continuous Deployment (CD)** by deploying the application on **Fly.io** with a **Supabase** database, utilizing **GitHub workflows** for seamless deployment and integration.

BOLT4 Onion Routing Implementation

March 2025

- Built a **Rust** implementation of the **BOLT-4 Onion Routing Protocol** from the Lightning Network for educational exploration.
- Implemented parsing of onion routing packets and extraction of payload data as per protocol specifications.
- Developed tooling to serialize and write onion packet data into hex format for inspection and debugging.
- Strengthened understanding of **cryptography**, **network protocols**, and **Bitcoin Lightning internals**.

HERMES – P2P Messaging & File Sharing

2025 – Present

- Contributing to a decentralized **peer-to-peer (P2P)** system aimed at replacing legacy DC++ on campus networks.
- Working on a cross-platform desktop client built with **Tauri (Rust)** and **Svelte (TypeScript)**.
- Leveraging **iroh** for networking, enabling efficient peer discovery, NAT traversal, and direct file transfers.
- Building real-time messaging and high-speed file sharing features with a focus on performance and security.

⚙️ SKILLS

- **Programming Languages:** Go, Rust, Typescript/Javascript, Bash, Python, C, Solidity
- **Frameworks / Libraries:** Svelte, ReactJS, HTMX, Astro, Next.js, Jekyll Django, Vue.js
- **Databases / Technologies:** SQLite, PostgreSQL, MongoDB, Node, Supabase
- **Systems / Platforms:** Linux, Git, Docker, Nix, Azure, CI/CD tools